

Year 11

Mathematics

Assessment Task 3

2017

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- In Questions 11 - 14, show relevant mathematical reasoning and/or calculations
- A reference sheet will be provided.

Total marks – 70

Section I

10 marks

- Attempt Questions 1 – 10
- Allow about 15 minutes for this section.

Section II

60 marks

- Attempt Questions 11-14
- Start each question in a new writing booklet
- Write your name on the front cover of each booklet to be handed in
- If you do not attempt a question, submit a blank booklet marked with your name and "N/A" on the front cover
- Allow about 1 hour 45 minutes for this section

DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

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Section I

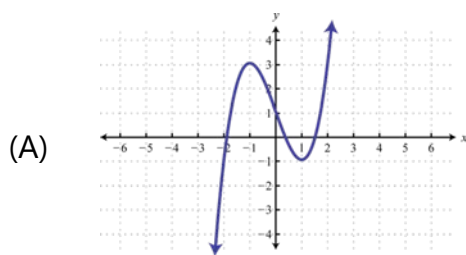
10 marks

Attempt Questions 1–10

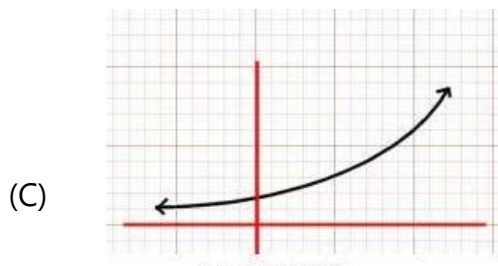
Allow about 15 minutes for this section

Use the multiple-choice answer sheet for Questions 1–10.

1 Which of the following is **not** a function



(B)



(D)

2 Rationalise $\frac{\sqrt{5}}{1+\sqrt{3}}$

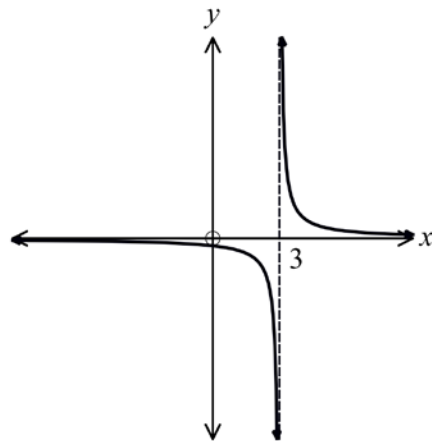
(A) $\frac{\sqrt{3}}{3}$

(B) $\frac{\sqrt{15}-\sqrt{5}}{2}$

(C) $\sqrt{5}$

(D) $\frac{\sqrt{5}+\sqrt{3}}{4+2\sqrt{3}}$

3 Which equation could describe the curve shown below



(A) $y = \frac{1}{x-9}$

(B) $y = \frac{1}{x-3}$

(C) $y = \frac{1}{x+3}$

(D) $y = \frac{1}{x+9}$

4 Solve $\sin \theta = \frac{1}{\sqrt{2}}$, $0 \leq \theta \leq 360$

(A) $\theta = 45^\circ, 135^\circ$

(B) $\theta = 135^\circ, 225^\circ$

(C) $\theta = 225^\circ, 315^\circ$

(D) $\theta = 45^\circ, 315^\circ$

5 Factorise $4m^2 - 49$

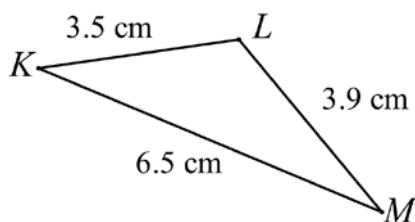
(A) $(2m+7)(2m-7)$

(B) $2(m+7)(m-7)$

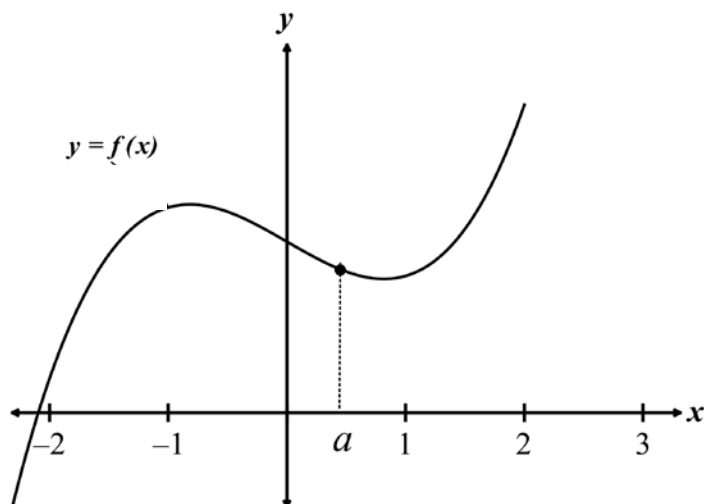
(C) $(2m-7)^2$

(D) $4(m-7)(m+7)$

- 6 Find the size of $\angle LKM$ to the nearest degree.



- (A) 28°
(B) 30°
(C) 35°
(D) 59°
- 7 The diagram shows part of the graph $y = f(x)$



Which of the following is a true statement?

- (A) $f(a)$ is positive and $f'(a)$ is positive
(B) $f(a)$ is negative and $f'(a)$ is positive
(C) $f(a)$ is positive and $f'(a)$ is negative
(D) $f(a)$ is negative and $f'(a)$ is negative
- 8 Which of the following is the domain of the function $y = \sqrt{x+4}$
- (A) $y \geq -0$
(B) $y > 0$
(C) $x \geq -4$
(D) $x > -4$

- 9** $A(3, y)$ and $B(7, 2)$ are points on the number plane. The gradient of AB is $m = -3$

What is the value of y ?

- (A) 14
- (B) $\frac{1}{14}$
- (C) -10
- (D) $-\frac{1}{10}$

- 10** Which of the following is equivalent to $\sqrt{3} + \sqrt{27} - \sqrt{18}$?

- (A) $\sqrt{3}$
- (B) $\sqrt{12}$
- (C) $4\sqrt{3} - 3\sqrt{2}$
- (D) $10\sqrt{3} - 9\sqrt{2}$

End of Section I

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Answer each question in the appropriate writing booklet. Extra writing booklets are available.

In Questions 11–14, your responses should include relevant mathematical reasoning and/or calculations.

Question 11 (15 marks)

Marks

(a) Solve each of the following equations

(i) $3x^2 + 2x - 5 = 0$ 2

(ii) $|2x - 5| - 1 = 0$ 2

(iii) $x^2 + 6x - 10 = 0$ 2

(b) Solve $3\tan^2 \theta = 1$ for $-180 \leq \theta \leq 180$ 2

(c) Prove each of the following identities:

(i) $\frac{1 - \sin^2 \theta}{\sin^2 \theta + \cos^2 \theta} = \cos^2 \theta$ 2

(ii) $\tan A \sin A + \cos A = \sec A$ 2

(d) Sapphire cycles 15 km due North, then 12 km due east and finally 20 km due south.

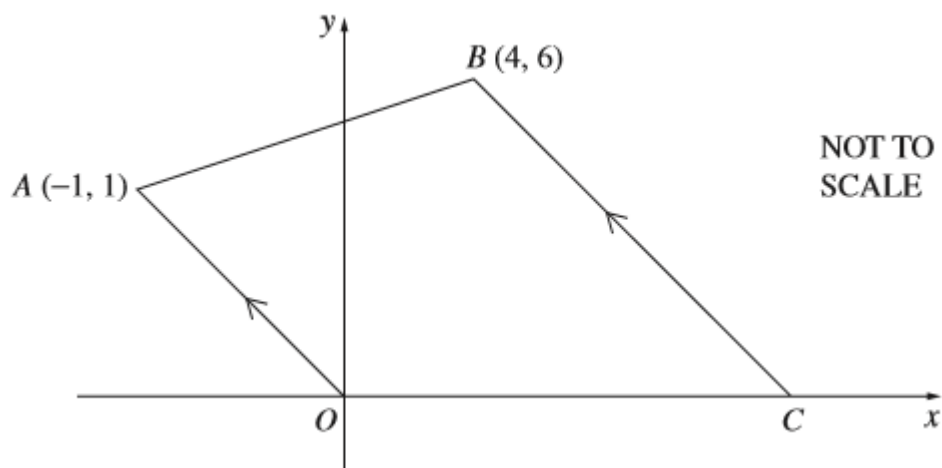
(i) Draw a diagram representing this information. 1

(ii) Determine her bearing to **(to the nearest degree)** and distance from her original position. 2

End of question 11

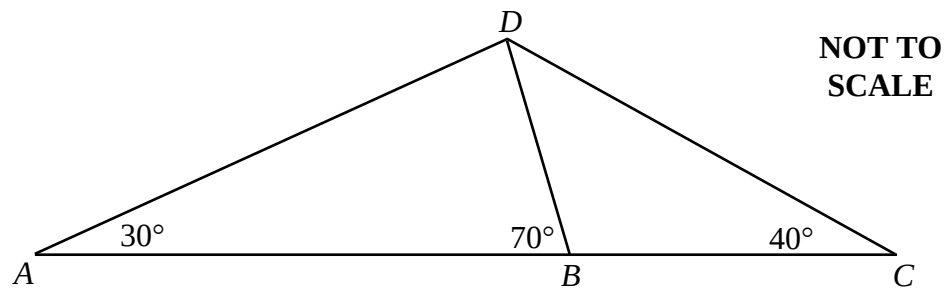
Question 12 (15 marks)**Marks**

- (a) In the diagram $OABC$ is a trapezium with $OA \parallel CB$. The coordinates of O , A and B are $(0, 0)$, $(-1, 1)$ and $(4, 6)$ respectively.



- | | | |
|-------|---|----------|
| (i) | Calculate the length of the interval OA | 1 |
| (ii) | Find the gradient of the line OA | 1 |
| (iii) | Find the equation of the line BC | 1 |
| (iv) | Hence, find the coordinates of the point C | 1 |
| (v) | Show that the perpendicular distance from O to the line BC is $5\sqrt{2}$ | 2 |
| (vi) | Calculate the area of the trapezium $OABC$ | 2 |
- b) Find the equation of the line that is parallel to $2x - y + 3 = 0$ and goes through the point $(2, 10)$
- 2**

- c) The diagram shows $\triangle ACD$ where $\angle DAC = 30^\circ$ and $\angle DCA = 40^\circ$. The point B lies on AC such that $\angle DBA = 70^\circ$.



- (i) Prove that $\triangle ACD$ and $\triangle DCB$ are similar.

3

Given that $CD = x$ cm, $AC = (x + 3)$ cm and $BC = (x - 2)$ cm, Using (i), or otherwise:

- (ii) Find the value of x .

2

End of question 12

Question 13 (15 marks)**Marks**

- (a) Show that the function $f(x) = 3x^4 - 4x^2$ is even. **2**
- (b) Find the equation of the circle with radius 3 units and centre $(0, -4)$. **1**
- (c) State the domain and range of the graph of $y = \frac{1}{2x-12}$. **2**
- (d) (i) Sketch the graph of $y = x^2 - 9$, $-4 \leq x \leq 3$ **2**
(ii) Hence determine its range **1**
- (e) Sketch $y = |2x - 6|$ **2**
- (f) Consider the functions $y = 6x - x^2$ and $y = 2x$
- (i) Find their points of intersection. **2**
- (ii) Shade the region defined by $y \leq 6x - x^2$ and $y > 2x$. **3**

End of question 13

Question 14 (15 marks)**Marks**

(a) Find the derivative function for $f(x) = x^2 - 3x$ using first principles.

2

(b) Differentiate each of the following

(i) $f(x) = x^2 - 10$

1

(ii) $f(x) = 3x^4 - 5x + 3$

1

(iii) $f(x) = \frac{x^3 - 6x^2}{x}$

1

(c) Consider the function $f(x) = \frac{1}{2}x^4 - 16x + 2$. Evaluate:

(i) $f(1)$

1

(ii) $f'(1)$

2

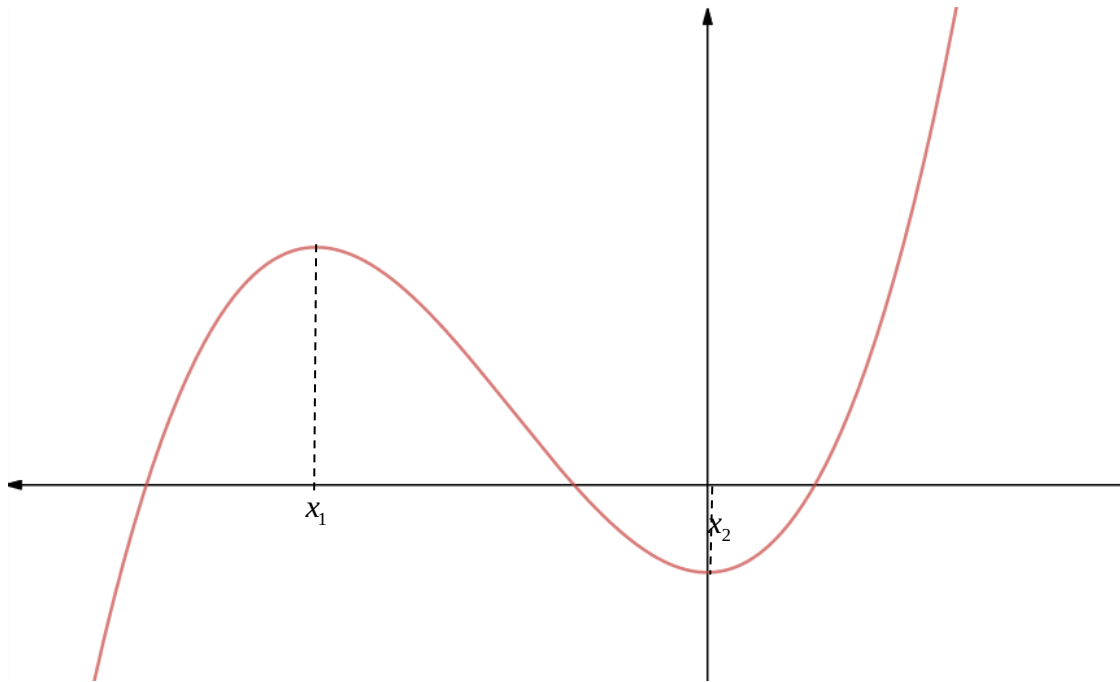
(iii) Determine any points on the curve at which the gradient is 0.

2

- (d) Copy or trace the diagram below into your answer booklet.

Sketch the gradient function for the following function

1



- (e) Consider $y = \frac{x^2 - 16}{x - 4}$

- (i) Find the limit:

2

$$\lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$$

- (ii) Hence sketch the curve $y = \frac{x^2 - 16}{x - 4}$, clearly showing the point of discontinuity.

2

End of Examination

Student Name:

2017 Year 11 Assessment Task 3

Mathematics

Section 1 Multiple-Choice Answer Sheet

- | | | | | |
|-----------|-------------------------|-------------------------|-------------------------|-------------------------|
| 1 | A ☐ | B ☐ | C ☐ | D ☐ |
| 2 | A ☐ | B ☐ | C ☐ | D ☐ |
| 3 | A ☐ | B ☐ | C ☐ | D ☐ |
| 4 | A ☐ | B ☐ | C ☐ | D ☐ |
| 5 | A ☐ | B ☐ | C ☐ | D ☐ |
| 6 | A ☐ | B ☐ | C ☐ | D ☐ |
| 7 | A ☐ | B ☐ | C ☐ | D ☐ |
| 8 | A ☐ | B ☐ | C ☐ | D ☐ |
| 9 | A ☐ | B ☐ | C ☐ | D ☐ |
| 10 | A ☐ | B ☐ | C ☐ | D ☐ |

Student Name: *Solutions*

2017 Year 11 Assessment Task 3

Mathematics

Section 1 Multiple-Choice Answer Sheet

- | | | | | |
|----|------------------------------------|------------------------------------|------------------------------------|-------------------------|
| 1 | A ☐ | B ☒ | C ☐ | D ☐ |
| 2 | A ☐ | B ☒ | C ☐ | D ☐ |
| 3 | A ☐ | B ☒ | C ☐ | D ☐ |
| 4 | A ☒ | B ☐ | C ☐ | D ☐ |
| 5 | A ☒ | B ☐ | C ☐ | D ☐ |
| 6 | A ☐ | B ☒ | C ☐ | D ☐ |
| 7 | A ☐ | B ☐ | C ☒ | D ☐ |
| 8 | A ☐ | B ☐ | C ☒ | D ☐ |
| 9 | A ☒ | B ☐ | C ☐ | D ☐ |
| 10 | A ☐ | B ☐ | C ☒ | D ☐ |

Question 11

a) i) $3x^2 + 2x - 5 = 0$ $P = -15$ $S = 2$
 $3x^2 - 3x + 5x - 5 = 0$
 $3x(x-1) + 5(x-1) = 0$ ✓ program
 $(3x+5)(x-1) = 0$
 $3x+5=0$ $x-1=0$
 $x = -\frac{5}{3}$ $x=1$ ✓ getting both.

ii) $|2x-5| = 1$
 $2x-5 = \pm 1$ ✓
 $2x-5 = 1$ $2x-5 = -1$
 $2x = 6$ $2x = 4$
 $x = 3$ $x = 2$ ✓ both

iii) $x^2 + 6x - 10 = 0$
 use quadratic formula OR
 Completing the square

$$x = \frac{-6 \pm \sqrt{6^2 - 4 \times 1 \times (-10)}}{2 \times 1}$$

$$= \frac{-6 \pm \sqrt{36 + 40}}{2}$$

$$= \frac{-6 \pm 2\sqrt{19}}{2}$$

$$x = -3 \pm \sqrt{19}$$
 ✓

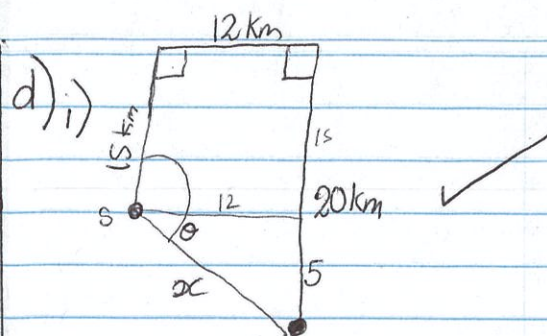
b) $\tan^2 \theta = \frac{1}{3}$ $\tan \theta = \pm \frac{1}{\sqrt{3}}$ ✓
 $\theta = \pm 30^\circ, \pm 150^\circ$ ✓
 ✓ 2 angles
 ✓ all 4 angles

c) i) $\frac{1 - \sin^2 \theta}{\sin^2 \theta + \cos^2 \theta} = \frac{\cos^2 \theta}{1} = \cos^2 \theta = \text{RHS}$ ✓

ii) $\tan A \sin A + \cos A = \frac{\sin A \times \sin A}{\cos A} + \cos A$

$$= \frac{\sin^2 A + \cos^2 A}{\cos A}$$
 ✓

$$= \frac{1}{\cos A} = \sec A$$
 ✓



d) i) $x^2 = 12^2 + 5^2$
 $x = \sqrt{12^2 + 5^2}$
 $x = 13 \text{ km}$ ✓

bearing $\rightarrow \tan \theta = \frac{5}{12}$

$$\theta = 23^\circ \text{ (nearest degree)}$$

$\therefore$ her bearing is $90 + 23$
 $= 113^\circ \text{ T}$ ✓

Question 12

i) $d = \sqrt{(0 - (-1))^2 + (6 - 1)^2}$
 $= \sqrt{1 + 25}$
 $= \sqrt{26} \text{ units}$ ✓

ii) $m_{OA} = \frac{1-0}{-1-0} = \frac{1}{-1} = -1$ ✓

iii) $m_{BC} = m_{OA} = -1$ $B(4, 6)$
 $y - 6 = -1(x - 4)$
 $y - 6 = -x + 4$
 $y = -x + 10$ ✓ OR $x + y - 10 = 0$

iv) at $y = 0 \rightarrow -x + 10 = 0$
 $x = 10$

$\therefore C(10, 0)$ ✓

vi) $BC = \sqrt{6^2 + 6^2} = \sqrt{72} = 6\sqrt{2}$ ✓
 $A_{OABC} = \frac{1}{2} \times 5\sqrt{2} (6\sqrt{2} + \sqrt{2})$
 $= 35 \text{ units}^2$ ✓

vii) $m = 2$ ✓ $y - 10 = 2(x - 2)$
 $y - 10 = 2x - 4 \rightarrow y = 2x + 6$ ✓

c) i) In $\triangle ACD$ and $\triangle DCB$
 $\angle PBC = 180 - 70 = 110^\circ$ ($\angle$'s on a straight line)
 $\therefore \angle BDC = 180 - (40 + 110)$ ($\angle$ sum of $\triangle$)
 $= 30^\circ$
 $\therefore \angle BDC = \angle DAC = 30^\circ$ (Proved above) ✓
 $\angle BCB = \angle DCA = 40^\circ$ (given) ✓
 $\therefore \angle DBC = \angle ADC$ ($\angle$ sum of $\triangle$) ✓
 $\therefore \triangle ACD \parallel \triangle DCB$ (AAA)

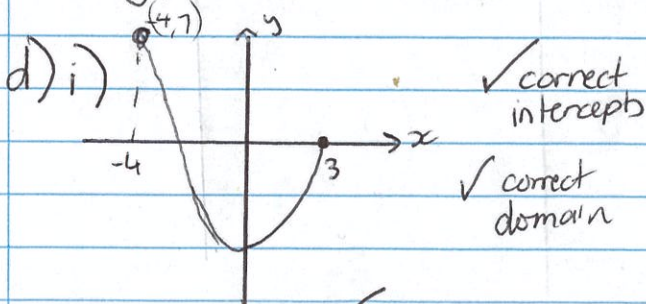
ii) $\frac{x}{x-2} = \frac{x+3}{x}$ ✓
 $x^2 = (x+3)(x-2)$
 $x^2 = x^2 + x - 6$
 $x - 6 = 0$
 $x = 6$ ✓

Question 13

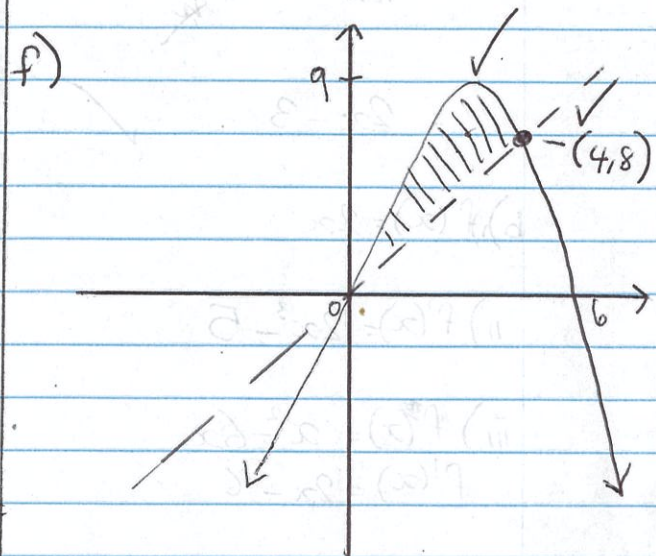
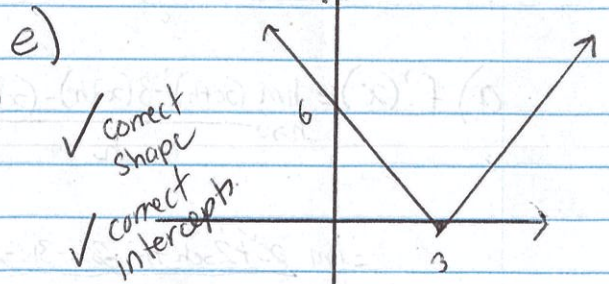
a) function is even if $f(x) = f(-x)$ ✓
 $f(x) = 3x^4 - 4x^2$
 $f(-x) = 3(-x)^4 - 4(-x)^2$
 $= 3x^4 - 4x^2$ ✓
 $\therefore f(x) = f(-x)$
Hence function is even

b) $x^2 + (y+4)^2 = 9$ ✓

c) D: $x \neq 6$ ✓
R: $y \neq 0$ ✓



ii) $-9 \leq y \leq 7$ ✓



ii) $6x - x^2 = 2x$
 $x^2 - 6x + 2x = 0$
 $x^2 - 4x = 0$
 $x(x-4) = 0$
 $x = 0 \quad x = 4$
 $y = 0 \quad y = 8$

POI $\rightarrow (0, 0)$ ✓ $(4, 8)$ ✓

iii) refer to diagram for shading ✓

Question 14

$$a) f'(x) = \lim_{h \rightarrow 0} \frac{(x+h)^2 - 3(x+h) - (x^2 - 3x)}{h}$$

$$= \lim_{h \rightarrow 0} \frac{x^2 + 2xh + h^2 - 3x - 3h - x^2 + 3x}{h}$$

$$= \lim_{h \rightarrow 0} \frac{2xh + h^2 - 3h}{h}$$

$$= 2x - 3$$

$$b) i) f'(x) = 2x$$

$$ii) f'(x) = 12x^3 - 5$$

$$iii) f(x) = x^2 - 6x$$

$$f'(x) = 2x - 6$$

$$c) f'(x) = 2x^3 - 16$$

$$i) f(1) = \frac{1}{2}(1)^4 - 16(1) + 2$$

$$= -13\frac{1}{2} \text{ OR } -27\frac{1}{2}$$

$$ii) f'(1) = 2(1)^3 - 16$$

$$= -14$$

$$iii) 2x^3 - 16 = 0$$

$$2x^3 = 16$$

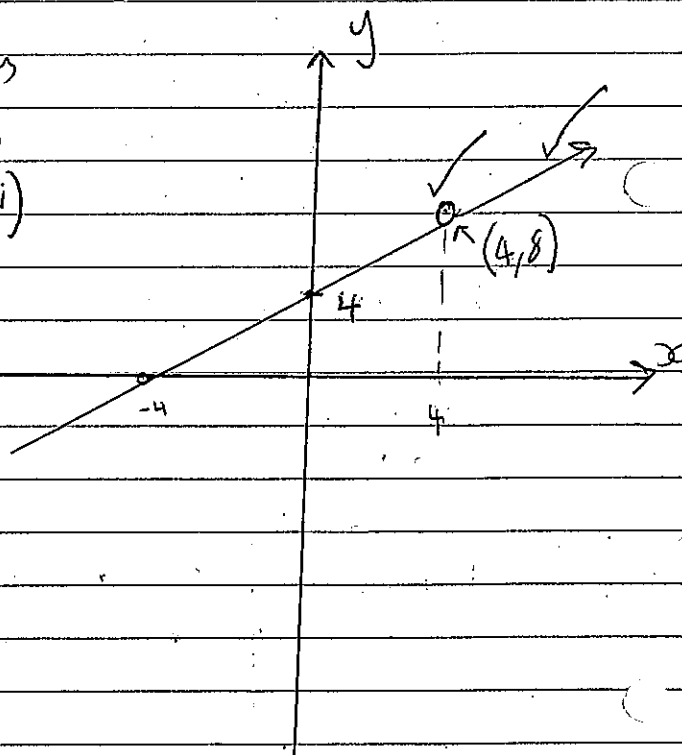
$$x^3 = 8$$

$$x = 2$$

$$e) i) \lim_{x \rightarrow 4} \frac{x^2 - 16}{x - 4}$$

$$= \lim_{x \rightarrow 4} \frac{(x-4)(x+4)}{x-4}$$

$$= 8$$



for differentiating
from the gradient
function